

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview

Objective

Develop an AI-based flood prediction system that provides early and accurate flood risk predictions using historical weather data.

Scope

The project analyzes rainfall and weather parameters using machine learning to predict flood occurrence through a user-friendly web application.

Problem Statement

Description

Existing flood warning systems often provide delayed or less accurate predictions, leading to property damage and loss of lives.

Impact

An accurate prediction system helps authorities and citizens take preventive actions, reducing flood damage and improving disaster preparedness.

Proposed Solution

Approach

Train a machine learning model using historical flood and rainfall data and deploy it as a Flask web application for real-time flood prediction.

Key Features

- Machine learning-based flood prediction
- Simple web interface for user input

- Fast and accurate prediction results
- Easy deployment and accessibility
- Scalable for future weather parameters

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i5/i7 or equivalent
	Memory	8 GB RAM
	Storage	256 GB SSD or above
Software	Framework	Flask
	Libraries	Scikit-learn, Pandas, NumPy, Joblib, OpenPyXL
	Development Environment	Visual Studio Code, Jupyter Notebook
Data	Source, Size, Format	Historical Flood Dataset (.xlsx) containing rainfall and weather data